Wind turbine project
Part I: Wave Dynamics
B. Estimating the extreme wave conditions
October 16, 2026

Following up on Part IA of your project, which consisted of characterizing the mean wave conditions and seasonal variability, today's class, Part IB, is focused on estimating the extreme wave conditions.

B. Estimating the extreme wave conditions

The design of marine structures requires knowledge about both the normal operating conditions as well as the extreme or abnormal environmental conditions to ensure the survival of the structure. A series of design load cases (or DLCs) exist in industry standards, which speify the technical requirements and provide guidance for how to estimate the loads and site conditions for wind turbines [1, 2].

To estimate the extreme wave conditions that may impact an offshore structure, it is necessary to complete an extreme value analysis (EVA) using long-term time series of wave conditions, such as the example shown in Figure 1. The results depend on the input data used in the statistical analysis, so the choice of data set (length and source, observations or modeling results). There may be large uncertainties in the EVA results depending on the data limitations, assumption of statistical stationarity (e.g. statistics of the wave climate are not changing), and inherent uncertainties in the EVA approach.

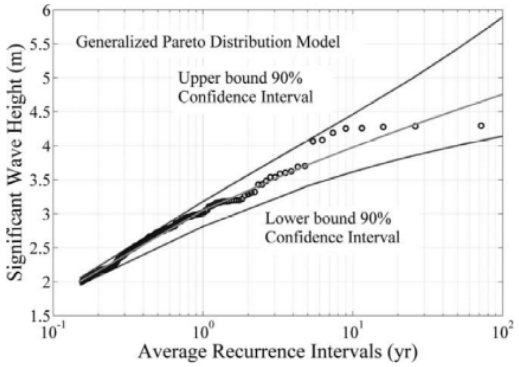


FIGURE 1. Example EVA analysis for the significant wave height off the coast of Malaysia (Figure 5, [3]).

For your project, you will need to evaluate the 1 and 50-year return period significant wave height for your study site by completing a univariate EVA of a long-term wave height time series. It is recommended that you use the data from the ResourceCode data base (or another similarly long time series) to have a time series of sufficient length to estimate the 50-year return period.

Block Maxima + GEVD. Calculate the estimated 1 and 50-year return periods using the block maxima method.

Q: Show a graphic of the results of your analysis and create a table with the 1 and 50-year return period values.

Peaks Over Threshold (POT). Calculate the estimated 1 and 50-year return periods using the POT method.

Q: Show a graphic of the results of your analysis, add the 1 and 50-year return period values to the previous table, and compare the results you obtained with the BM and POT methods. How similar are your results?

Estimating the associated wave period and direction. To estimate the associated peak wave period, calculate the 10, 50 and 90% quantile peak wave periods associated with the extreme wave events selected in the previous sections. Finally, for the wave direction data, estimate the mean wave direction and spreading using the entire time series of observations.

Q: Summarize the results of your analyses in a table. Is the mean wave direction representative of the observed wave direction variability (from the previous class)?

5. (optional) Comparison to multivariate method. If you have time to go further, you may search for the MHKiT python toolbox that will allow you to calculate the 50-year return contour using the modified I-FORM method.

Q: Show a graphic of your results and compare this to the values obtained with the univariate analysis (e.g. add those points to the environmental contour plot). Are they similar?

Your first inidividual, intermediary project report summarizing your work in parts IA and IB is due on October 23, 2026. Please deposit the file in the following folder via this link: https://cloud.enpc.fr/s/B8MQJsigGxLM4ni
Please include the requested figures and responses to the questions asked here and in class last week, and include any additional relevant data interpretation. Be clear and concise.

REFERENCES

[1] DNV, Support structures for wind turbines, Standard, DNV (2021).

[2] IEC, Wind energy generation systems - Part 3-1: Design requirements for fixed offshore wind turbines, International standard, IEC (2019).

[3] S. S. Far, A. K. A. Wahab, S. B. Harun, Determination of significant wave height offshore of the federal territory of labuan (malaysia) using generalized pareto distribution method, Journal of Coastal Research 34 (4) (2018) 892–899.